

User Manual: A.R.T.E. System & Modules

This manual describes the correct startup and initialization of the A.R.T.E. System, how to load the associated modules, and the fundamental guidelines for new development.

1. System Startup (Strict Sequence)

The A.R.T.E. System requires a strict boot sequence. The core system must **always be loaded first**.

1. Navigate to the `./System` folder on your computer.
2. Open the file `ARTE_System_Core.html` in a modern web browser (e.g., by double-clicking the file).
3. Wait until the "Open Origin - OS Control Room" dashboard has fully loaded and the status displays "Ready".

2. Loading Modules

Once the system core is running, the desired modules from the `./Module` folder must be injected into the system. You have two options for this:

Method A: Drag & Drop (Recommended)

1. Open your file explorer and navigate to the `./Module` folder.
2. Select the desired HTML module files.
3. Drag and drop the files directly into the dashed "Drop Zone" in the ARTE Core Dashboard (labeled "Drag & drop modules here").

Method B: Manual Upload

1. Click on the "Drop Zone" in the ARTE Core Dashboard.
2. A file selection dialog will open.
3. Navigate to the `./Module` folder, select the desired HTML files, and confirm the upload.

The system will now read the modules and make them available as new, interactive tabs in the "Workspace Level" (bottom tab bar).

3. Overview of Available Modules

The following modules can be loaded and used after the system startup:

- **Audio_Analyser-Pipeline.html (Acoustic Resonance Engine):** Visualizes audio files in real-time with various reactive effects (e.g., Cyberpunk Matrix, Nebula).
- **Image_Resizer_Matrix.html (High Fidelity Scaler):** Allows precise scaling of images while maintaining aspect ratio and provides lossless export to various formats (PNG, JPG, WEBP).
- **KineticTelemetry&_Canvas_Engine.html:** A GPU-accelerated oscilloscope for real-time data visualization on an HTML5 canvas.

- **OpticalContour&_Blueprint_Synthesizer.html**: Converts uploaded images locally via edge detection into detailed blueprints or sketches.
 - **ScreenCapture&_Canvas_Interaction.html (FAKE-RISK ENGINE)**: An advanced analysis tool for checking image or video data for manipulation utilizing blacklists, NLP patterns, and OCR.
 - **Kinetic Capture & Imaging Engine**: Couples the system with video feeds or screens to create snapshots, video recordings, and perform direct image editing in a focus-room studio.
 - **Advanced Document Synthesis Matrix**: A local WYSIWYG text editor featuring A4 paper simulation, image drag & drop, and export options (PDF, HTML, MD, TXT).
 - **Universal Media Converter (Universal Media Transcoder)**: Transcodes media entirely offline within the browser. Extracts audio tracks for high-quality MP3 export (via integrated LameJS) and native formats (MP4, WebM, M4A).
 - **Kinetic Video Engine**: Allows loading videos for real-time playback, precise cropping, and exporting selected segments into adjustable codecs.
 - **Global Weather Radar / 48H Forecast**: An interactive, global weather map including radar animations, toggleable weather layers, and a 48-hour forecast.
 - **Live Web Synthesis Matrix**: An IDE (Integrated Development Environment) directly in the browser for HTML/CSS/JS with live preview capabilities and code export.
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4. Guidelines for Developing New Modules

When developing new A.R.T.E. modules, the following architectural and content standards must be strictly adhered to in order to ensure system stability and alignment:

- **Shadow DOM Isolation**: For security reasons, every module is loaded by the core system (`ARTE_System_Core.html`) into its own isolated Shadow DOM. Global queries like `document.getElementById()` will fail. A dynamic root reference must always be established (e.g., `const root = typeof shadowRoot !== 'undefined' ? shadowRoot : document;`).
- **Local i18n & Event Bus (`osBus`)**: Modules must function entirely bi-lingually (German/English) and translate themselves natively. A local dictionary object is required. Furthermore, the module must listen to the global event bus for language changes (`window.osBus.on('lang-changed', ...)`) to dynamically update the user interface without triggering a page reload.
- **Scientific Precision & Functionality**: Modules must possess a clearly functional character. Any calculations, interpretations, and processes must be constructed to be mathematically precise, comprehensible, and consistent with physical and chemical foundations. Unverified assumptions or speculative logical leaps must be avoided.
- **Universal Utility & Strict Neutrality**: Written code, user interfaces, and documentation must be designed to offer broad, general utility. The inclusion of any political statements, judgments, or biases—regardless of language or project scope—is strictly prohibited without exception.